
資料視覚化

Data Visualization

Week 11: Topic Models

Roadmap for today

1. Embeddings

Why text must be converted into numbers

How similar documents end up close together in vector space

Why semantic similarity matters for topic modelling

2. Clustering

How we group nearby documents into candidate topics

Difference between dense regions and isolated points

Why not every document belongs cleanly to one topic

3. BERTopic

Embedding model + UMAP + HDBSCAN + c-TF-IDF

Train on 20 Newsgroups

Interpret topics with keywords and representative documents

4. Validation

Check whether topics are meaningful for the research question

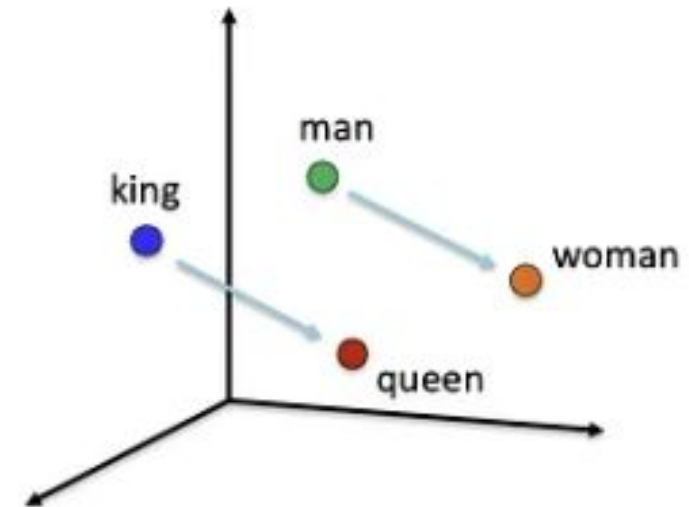
Use close reading, stability checks, and external evidence

Why validation matters more than pretty charts

What are word embeddings?

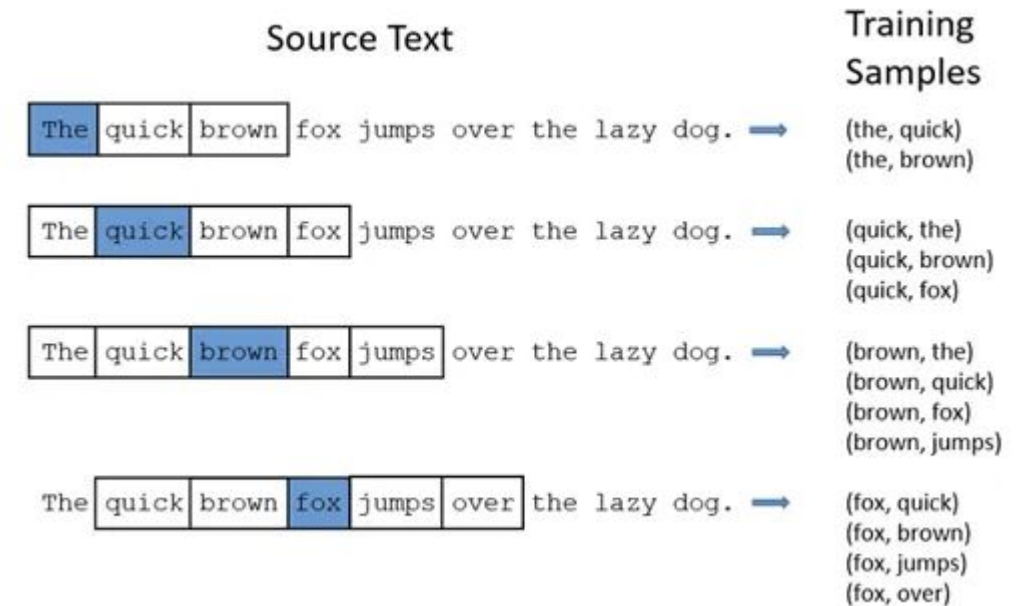
- A technique to represent words as numerical vectors in a high-dimensional space
- Words with similar meaning should have similar numbers
- Can be projected onto a 2D plane (known as dimension reduction)
- You can calculate their similarity using cosine (still remember that from high school?)
- You can get document embeddings from word embeddings

Words	Word Embeddings
"queen"	[0.1 0.2 0.7 0.3 0.2 0.8]
"king"	[0.8 0.5 0.1 0.9 0.7 0.2]
"man"	[0.5 0.6 0.3 0.2 0.4 0.1]
"woman"	[0.9 0.8 0.4 0.1 0.1 0.2]



Word embeddings models

- We used to train word embeddings models ourselves
- Simple intuition: words that often appear together should have similar meaning, vice versa
- We don't do that anymore, most of us now piggy-back on pre-train models, like BERT



Source: <https://medium.com/@zafaralibagh6/simple-tutorial-on-word-embedding-and-word2vec-43d477624b6d>

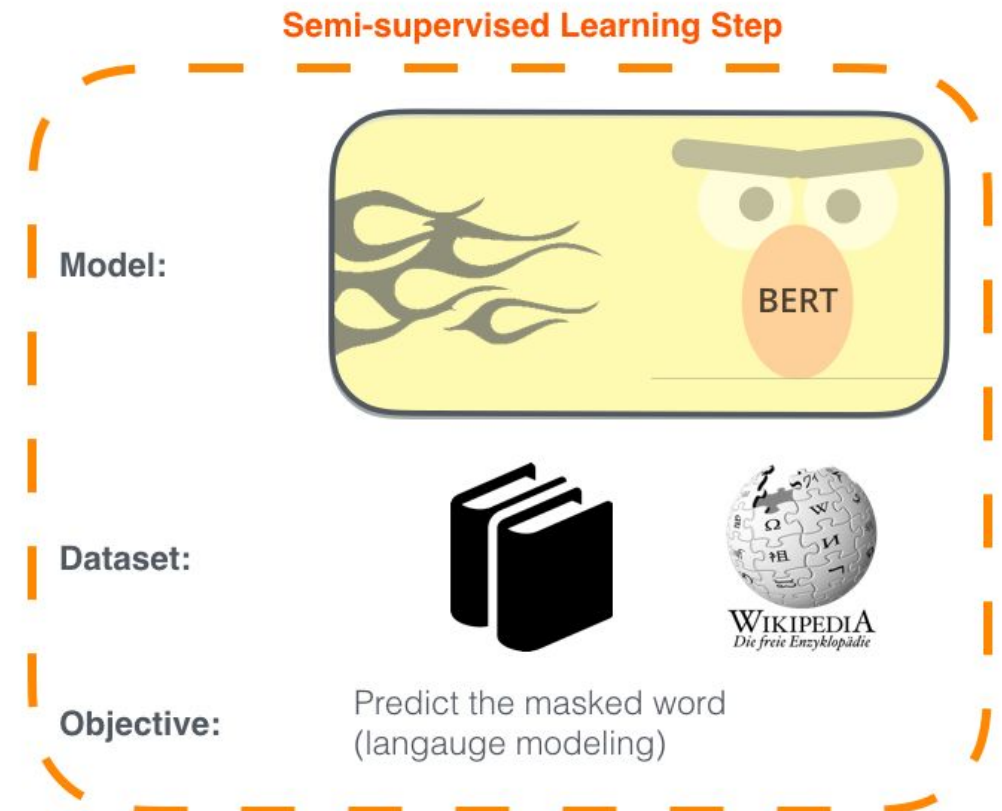
BERT

(Bidirectional Encoder Representations from Transformers)

- the OG Large Language Model
- It “learn” the language by training on a massive amount of texts
- It can support task such as machine learning classification, translation (google translate), and text generation
- We can use it to turn words and documents into embeddings

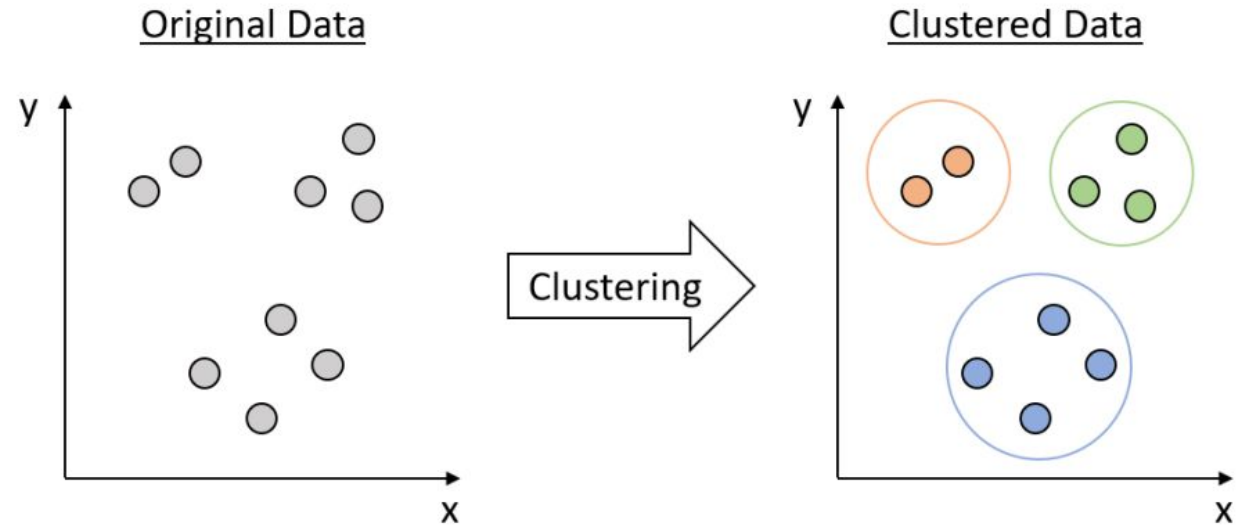
1 - **Semi-supervised** training on large amounts of text (books, wikipedia..etc).

The model is trained on a certain task that enables it to grasp patterns in language. By the end of the training process, BERT has language-processing abilities capable of empowering many models we later need to build and train in a supervised way.



Clustering

- Clustering is a technique to put data into groups
- In BERTopic, we put documents into groups of documents based on their meaning
- Note that I am skipping a lot of technique details here



Source: <https://www.linkedin.com/pulse/what-clustering-machine-learning-avishek-patra-ap>

BERTopic

- BERTopic is a package that do the embedding and clustering for you
- It will also generate a “representation” from each group, so that you know what they are able
- The basic pipeline:



Basic BERTopic Workflow



BERTopic

fit the pipeline



Interpret

name and inspect



Validate

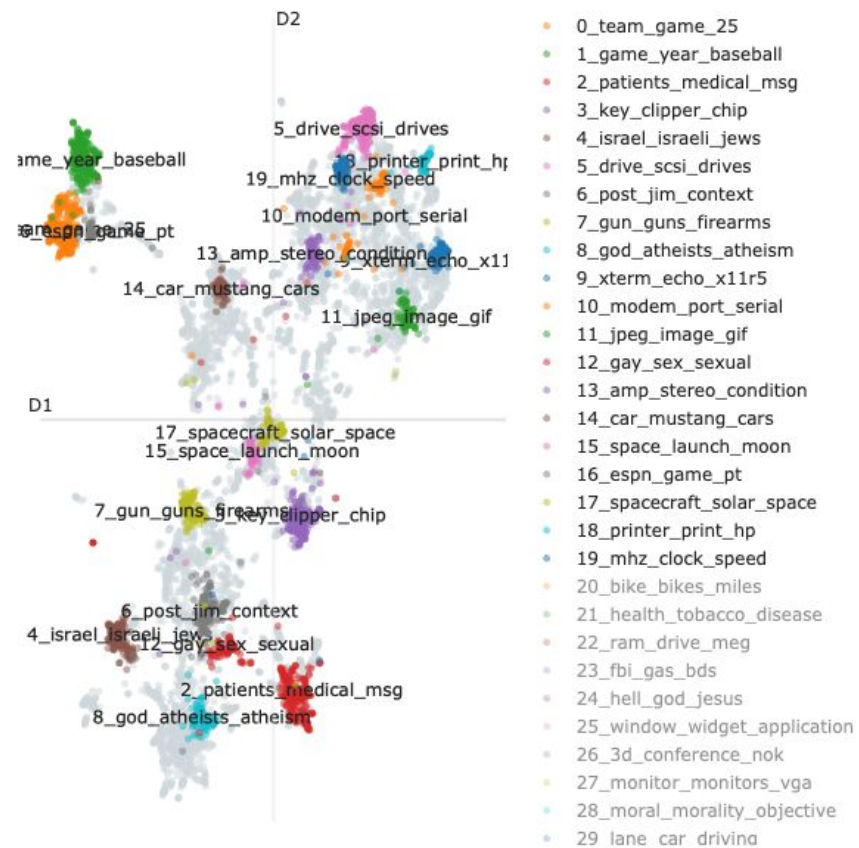
check usefulness

Interpret the Topics

Topic Word Scores



Distance between Topics



Validate, Validate, Validate

- Topic models can produce topics that look good on the surface, but incoherent when you take a closer look
- Labels can hide ambiguity, mixed themes, or artifacts from preprocessing
- Validation checks whether the output is useful, trustworthy, and fit for purpose
- There are multiple ways to validate, the easiest way is human annotation
- Rule of thumb:
It aren't a good topic model if it cannot be replicated by a human

Applying LDA Topic Modeling in Communication Research: Toward a Valid and Reliable Methodology

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ABSTRACT

Latent Dirichlet allocation (LDA) topic models are increasingly being used in communication research. Yet, questions regarding reliability and validity of the approach have received little attention thus far. In applying LDA to textual data, researchers need to tackle at least four major challenges that affect these criteria: (a) appropriate pre-processing of the text collection; (b) adequate selection of model parameters, including the number of topics to be generated; (c) evaluation of the model's reliability; and (d) the process of validly interpreting the resulting topics. We review the research literature dealing with these questions and propose a methodology that approaches these challenges. Our overall goal is to make LDA topic modeling more accessible to communication researchers and to ensure compliance with disciplinary standards. Consequently, we develop a brief hands-on user guide for applying LDA topic modeling. We demonstrate the value of our approach with empirical data from an ongoing research project.

Introduction

Topic modeling with latent Dirichlet allocation (LDA) is a computational content-analysis technique that can be used to investigate the "hidden" thematic structure of a given collection of texts. The data-driven and computational nature of LDA makes it attractive for communication research because it allows for quickly and efficiently deriving the thematic structure of large amounts of text documents. It combines an inductive approach with quantitative measurements, making it particularly suitable for exploratory and descriptive analyses (Elgesem, Steskal, & Diakopoulos, 2015; Koltsova & Shcherbak, 2015).

Consequently, LDA topic models are increasingly being used in communication research. However, communication scholars have not yet developed good-practice guidance for the many challenges a user faces when applying LDA topic modeling. Important methodological decisions must be made that are rarely explained at length in application-focused studies. These decisions relate to at least four challenging questions: (a) How does one pre-process unstructured text data appropriately? (b) How does one select algorithm parameters appropriately, e.g., the number of topics to be generated? (c) How can one evaluate and, if necessary, improve reliability and interpretability of the model solution? (d) How can one validate the resulting topics?

These challenges particularly affect the approach's reliability and validity, both of which are core criteria for content analysis in communication research (Neuendorf, 2017), but they have, nevertheless, received little attention thus far. This article's aim is to provide a thorough review and

A photograph of a monkey sitting in a black office chair at a dark wooden desk. The monkey is wearing a light-colored, textured sweater and is typing on a silver laptop. The laptop screen is open and tilted back. The background is a wall with vertical wood paneling. The text "Time for Python" is overlaid in white, bold, sans-serif font across the middle of the image.

Time for Python